



Injector Modelling and Design

Analytical Mass Flow Prediction and CFD Validation



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1. Introduction

The activity described in this report is part of the ICARUS group development program and aims at the analytical, numerical, and CFD modeling of the fluid-dynamic behavior of injectors for hybrid rocket engines using nitrous oxide (N_2O) and paraffin.

In particular, the project addresses the need for reliable predictive tools capable of estimating the mass flow rate through single-orifice and multi-orifice configurations, assessing the influence of geometry on the discharge coefficient, and supporting the selection of the injector based on the mass flow requirements imposed by the propulsion system. Since the experimental realization and testing of multiple geometries would require significant time and cost, it is necessary to adopt an approach that combines theoretical modeling, low-dimensional numerical tools, and targeted CFD simulations.

A key objective of this work is the definition of a methodology to estimate the discharge coefficient C_d based on empirical formulations and geometric correction factors, primarily dependent on the inlet rounding ratio r/D and the length-to-diameter ratio L/D . Such estimation enables both the prediction of the resulting mass flow rate for a given geometry and pressure drop, and the identification of geometric configurations that, for a prescribed target mass flow rate, maximize the discharge coefficient. This approach makes it possible to select the most efficient injector in terms of losses and feeding capability.

The work is organized into two complementary macro-activities:

- **Performance.** Given as input the geometry, pressure drop, and thermodynamic conditions, the pipeline computes the mass flow rate using a phase-aware model based on saturation-condition analysis (SPI, HEM, and NHNE approaches). If a discharge coefficient is available from CFD or experiments, it can be used directly; otherwise, it is estimated through geometric correlations derived from the literature. The output is the mass flow rate associated with the specified configuration.
- **Design.** For a prescribed target mass flow rate and pressure drop, this module explores the geometric space $(D, r/D, L/D)$ and identifies the parameter combination that best reproduces the required mass flow, while prioritizing geometries with the highest discharge coefficient. This yields an injector siz-

ing methodology guided both by the mass flow objective and by fluid-dynamic efficiency.

To support these tasks, a dedicated Python suite has been developed, organized into three main modules:

- a **physical core** for the phase-aware mass flow evaluation;
- a **performance** module for parametric analysis and reconstruction of interpolated surfaces $C_d(r/D, L/D, \text{Re})$;
- a **design** module for identifying the optimal geometry under the imposed constraints.

The combination of these elements enables the construction of a robust, replicable, and easily extensible injector design process, reducing the dependence on preliminary physical testing and allowing rapid iteration over geometric choices. The report therefore provides a complete description of the methodology, the CFD validation campaign, and future developments, including multiphase simulations and the formulation of a dedicated model for swirl injectors.

2. Numerical modelling method

This chapter describes the numerical workflow developed for the estimation of mass flow rates and for the sizing of plain-orifice injectors fed with N_2O . The goal is to show how the three Python modules `PyInjection_core`, `PyInjection_performance` and `PyInjection_design` implement the theoretical models introduced in the previous chapter (SPI, HEM, NHNE and the C_d model), and how these are combined to support both the analysis of a single case and the inverse design of candidate geometries.

The text explicitly links:

- the numerical implementation with the basic physical models (phase-aware selection of the mass-flow model, reconstruction of the outlet state, geometric estimation of C_d);
- the robustness choices adopted near the saturation region of N_2O (safe extraction of properties, liquid/gas fallback, convergence criteria);
- the role of each module within the chain: *physical core* $\rightarrow$ *single-case performance analysis* $\rightarrow$ *search for optimal geometries*.

2.1 Overall structure of the Python code

The code architecture has been designed in a hierarchical way:

- the *core* module (`pyinjection_core`) contains all the physics of the problem, that is the mass-flow models (SPI, HEM, NHNE), the phase-aware selector, the reconstruction of the outlet state and the semi-empirical model of the discharge coefficient C_d ;
- the *performance* module (`pyinjection_performance`) provides a *single-case* interface that calls the core and organises the results in readable tables, intended for comparison with CFD cases or experimental measurements;
- the *design* module (`pyinjection_design`) implements an *inverse-design* tool, which explores a three-dimensional grid in the geometric space and identifies

the configurations that satisfy a target mass-flow requirement with the highest possible discharge coefficient.

All modules share the same thermodynamic backend, implemented through `CoolProp`. Pressures are internally converted to Pascal, while user interfaces typically work in bar; similarly, temperatures and thermophysical properties are always handled in SI units. Particular attention has been devoted to numerical robustness near the saturation region and critical conditions of N_2O , where the evaluation of properties can be sensitive to input conditions: for this reason additional checks have been introduced on the state (single-phase/two-phase), “safe” functions for density and viscosity, and limits on admissible values.

In the following paragraphs, each module is described in more detail: first the physical backend `pyinjection_core`, then the analysis `pyinjection_performance` and finally the inverse-design tool `pyinjection_design`.

2.2 `pyinjection_core` module

The file `pyinjection_core.py` constitutes the physical heart of the model. In addition to the three mass-flow models, it includes a set of utility functions for robust extraction of thermodynamic properties (saturation temperature, density, viscosity, speed of sound), capable of handling errors or inconsistencies of `CoolProp` near the saturation curve of N_2O . In particular, liquid/gas fallbacks and safety limits on density and viscosity are introduced, so as to avoid numerical divergences in limiting cases.

2.2.1 SPI, HEM and NHNE mass-flow models

The reference mass flow rate is always calculated for a single hole, assuming a pressure drop

$$\Delta p = P_1 - P_2$$

between the upstream manifold and the downstream chamber.

SPI model

The SPI model (*Single Phase Incompressible/Compressible*) represents the single-phase limit of the problem and is used by the phase-aware selector whenever the energy balance indicates a non-two-phase outlet.

The code first identifies the prevailing upstream phase by comparing the line temperature with the saturation temperature at pressure P_1 ,

$$T_{\text{line}} \gtrless T_{\text{sat}}(P_1),$$

and computes a reference density ρ through robust functions that avoid numerical instabilities near saturation.

The ideal incompressible mass flow rate is estimated as:

$$\dot{m}_{\text{SPI,inc}} = C_d A \sqrt{2 \rho \Delta p},$$

a formula widely used in studies on injectors for self-pressurising oxidisers such as N_2O .¹ Here A is the hole area and $\Delta p = P_1 - P_2$.

In the input interface of the **performance** module it is possible to activate a compressibility correction based on the dimensionless factor Y' :

$$\dot{m}_{\text{SPI}} = C_d A Y' \sqrt{2 \rho \Delta p}, \quad Y' = \frac{1}{\sqrt{1 + Y_p}}.$$

The code implements two alternative forms of Y_p , both derived from the literature on compressible liquids and on flow-rate measurements through orifices for real fluids.²

1. Polytropic form (if the user specifies exponent n)

$$Y_p = \frac{n}{n-1} \left[1 - \left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}} \right].$$

2. Form based on the liquid bulk modulus

$$Y_p = \frac{\Delta p}{K} = \frac{\Delta p}{\rho a^2},$$

where a is the speed of sound at the upstream condition.

The structure implemented in the code deliberately keeps a simplified form compared to the full expressions, so as to guarantee numerical stability even close to N_2O saturation.

When the correction is deactivated, the SPI model reverts to the incompressible formulation and represents the reference single-phase limit in the automatic selection of the mass-flow model.

HEM model

The HEM model (*Homogeneous Equilibrium Model*) represents the two-phase limit of the problem: it is assumed that the two phases move with the same velocity and that the mixture is in thermodynamic equilibrium. This approach is widely used in the literature on injectors for N_2O and in mass-flow models under flashing or

¹Vargas Niño and Razavi 2019; Waxman et al. 2019.

²Zimmerman et al. 2012; Perry and Green 1984.

controlled cavitation conditions.³

In the code an energetic version consistent with the classical formulation of Waxman et al.⁴ is implemented, in which the HEM mass flow rate is expressed as:

$$\dot{m}_{\text{HEM}} = C_d A \rho_2 \sqrt{2(h_1 - h_2)},$$

with the isentropic condition

$$s_1 = s_2.$$

Operationally, the `_mdot_hem` function performs the following main steps:

- **Initial checks and physical upper bound.** If the diameter D , discharge coefficient C_d or pressure drop $P_1 - P_2$ are not physically valid (zero/negative diameter, $C_d \leq 0$, $P_1 \leq P_2$), the function directly returns $\dot{m}_{\text{HEM}} = 0$. Before the actual calculation, an upper-limit mass flow rate based on the incompressible SPI model with the same C_d is evaluated:

$$\dot{m}_{\text{SPI,upper}} = \dot{m}_{\text{SPI,inc}}(C_d, A, \rho_{\text{ref}}, P_1, P_2),$$

which is then used as a final *physical clamp*: the HEM model is never allowed to exceed the corresponding incompressible single-phase mass flow.

- **Upstream state: enthalpy and entropy (h_1, s_1).** The upstream stagnation enthalpy is obtained through `CoolProp` starting from (P_1, T_{line}) :

$$h_1 = h(P_1, T_{\text{line}}), \quad s_1 = s(P_1, T_{\text{line}}).$$

In case of numerical error, the code falls back to saturated liquid at P_1 ($Q=0$), thus still providing consistent values of h_1 and s_1 .

- **Downstream isentropic state (P_2, s_1).** The outlet state is obtained by directly imposing the isentropic condition:

$$h_2 = h(P_2, s_1), \quad \rho_2 = \rho(P_2, s_1).$$

If the solution at (P_2, s_1) fails (for instance near the critical region), the code falls back to an isothermal estimate by evaluating (P_2, T_{line}) :

$$h_2 \approx h(P_2, T_{\text{line}}), \quad \rho_2 \approx \rho(P_2, T_{\text{line}}).$$

- **HEM mass-flow calculation.** Once h_1 , h_2 and ρ_2 are known, the useful enthalpy drop is

$$\Delta h = \max(h_1 - h_2, 0),$$

³Vargas Niño and Razavi 2019; Waxman et al. 2019.

⁴Waxman et al. 2019.

and the HEM mass flow rate is computed as

$$\dot{m}_{\text{HEM}} = C_d A \rho_2 \sqrt{2 \Delta h}.$$

The value is forced to be non-negative and, finally, limited from above by the incompressible SPI value:

$$\dot{m}_{\text{HEM}} \leftarrow \min(\max(\dot{m}_{\text{HEM}}, 0), \dot{m}_{\text{SPI,upper}}).$$

In this way, the HEM model implemented in the code provides a two-phase estimate consistent with the enthalpy balance between upstream and downstream and with the homogeneous-equilibrium assumption, directly exploiting Waxman's relation and the thermodynamic closure offered by `CoolProp`. In the overall workflow, the value of $\dot{m}_{\text{HEM}}$ thus obtained is used by the NHNE model to build a controlled transition between the single-phase regime (SPI) and the equilibrium two-phase regime (HEM).

NHNE model

The metastable behaviour of N_2O during a rapid expansion is well documented in experimental tests on single-hole injectors⁵ and in several analytical models developed to predict the two-phase mass flow rate in out-of-equilibrium conditions.⁶ To represent this intermediate regime, the code implements an NHNE model (*Non-Homogeneous Non-Equilibrium*) that interpolates in a controlled way between the single-phase limit (SPI) and the equilibrium two-phase limit (HEM).

The implemented formulation is:

$$\dot{m}_{\text{NHNE}} = \frac{k}{k+1} \dot{m}_{\text{SPI}} + \frac{1}{k+1} \dot{m}_{\text{HEM}},$$

where `_mdot_spi` and `_mdot_hem` provide respectively the two extreme physical limits, while the blending coefficient k is computed by the `_k_local` function as:

$$k = \sqrt{\frac{P_1 - P_2}{\max(P_{\text{sat}}(T_{\text{line}}) - P_2, P_\epsilon)}}.$$

This structure reflects two physical properties observed in the literature:

- when the upstream pressure P_1 is much higher than the saturation pressure at line temperature, the fluid remains largely subcooled and the single-phase behaviour dominates, so $k \gg 1$ and $\dot{m}_{\text{NHNE}} \approx \dot{m}_{\text{SPI}}$;

⁵Zimmerman et al. 2012; Waxman et al. 2019.

⁶Vargas Niño and Razavi 2019.

- when P_1 approaches the saturation curve, the flow becomes sensitive to two-phase phenomena and the HEM component gains weight, leading to $k \rightarrow 0$ and $\dot{m}_{\text{NHNE}} \approx \dot{m}_{\text{HEM}}$.

The `_mdot_nhne` function returns both the resulting mass flow rate and the value of k , which can be used for diagnostics or for visualisation of the operating regime.

2.2.2 Phase-aware selector

The high-level function `compute_mdot_phaseaware` is the main API for mass-flow estimation in the Python code.

Given a set of input parameters (fluid, pressures, line temperature, diameter and discharge coefficient), the selector hides the details of the individual models from the user and returns a single “physically plausible” mass flow rate together with the information about the actually used model.

From the interface point of view, `compute_mdot_phaseaware` receives as input:

- the fluid (`fluid`) and the upstream and downstream pressures in bar (`p1_bar`, `p2_bar`);
- the line temperature T_{line} (`T_line`);
- the orifice diameter D and discharge coefficient C_d (`D`, `Cd`);
- two options for the SPI model: whether to use the compressibility correction or not (`use_spi_compress`) and the optional polytropic exponent n (`spi_n`).

The internal logic follows three main steps:

1. **Evaluation of the mass-flow models.** The single-phase mass flow rate SPI is calculated:

$$\dot{m}_{\text{SPI}} = \text{_mdot_spi}(\dots),$$

and the intermediate NHNE mass flow rate is computed:

$$\dot{m}_{\text{NHNE}} = \text{_mdot_nhne}(\dots),$$

where NHNE combines SPI and HEM with weights according to the philosophy of non-equilibrium models for N_2O injected through short orifices.⁷

2. **Phase diagnosis with the SPI model.** An estimate of the expected outlet phase is then obtained based on the SPI solution via the auxiliary function `_phase_from_spi_guess`. This function:

- reconstructs the average outlet velocity U_{SPI} from $\dot{m}_{\text{SPI}}$;

⁷Zimmerman et al. 2012; Waxman et al. 2019; Vargas Niño and Razavi 2019.

- computes the upstream stagnation enthalpy h_1 and the downstream enthalpy $h_2 = h_1 - U_{\text{SPI}}^2/2$;
- compares h_2 with the saturated-liquid and saturated-vapour enthalpies at P_2 (obtained from CoolProp⁸) to decide whether the estimated state is liquid, gaseous or two-phase.

The output is a string ("liquid", "gas" or "two-phase") summarising the phase diagnosis of the SPI model.

3. **Automatic model selection.** If the diagnosis returns "two-phase", the selector assumes that the SPI model is no longer adequate to describe the two-phase relaxation in the orifice and replaces the solution with the NHNE mass flow rate:

$$\dot{m}_{\text{PA}} = \begin{cases} \dot{m}_{\text{NHNE}}, & \text{if the SPI estimate is two-phase,} \\ \dot{m}_{\text{SPI}}, & \text{otherwise.} \end{cases}$$

Otherwise, if the state predicted by SPI is single-phase (liquid or gas), the SPI solution is retained, as it represents the simplest and least dissipative limit.

The `compute_mdot_phaseaware` function finally returns a pair (`mdot_phaseaware`, `model_used`), where `mdot_phaseaware` is the selected mass flow rate and `model_used` is a string in {"SPI", "NHNE"}. This information is propagated to the `pyinjection_performance` and `pyinjection_design` modules, allowing one to track, for each analysed case or each explored geometry, which physical regime has actually been chosen by the phase-aware selector and under which assumptions (incompressible/compressible SPI or NHNE).

2.2.3 Reconstruction of the outlet state

To compare the 0D model with CFD simulations or experimental measurements, it is useful to reconstruct a thermodynamic state at the outlet section consistent with the computed mass flow rate. This operation is implemented by the function `nhne_out_state_from_mdot` which, given a value of $\dot{m}$ (typically the *phase-aware* mass flow rate) and the orifice geometry, reconstructs a complete block of properties at the outlet by assuming an isentropic expansion, in full consistency with the HEM model implemented earlier.

From a numerical standpoint, the function performs the following main steps:

1. **Initialisation and upstream state.** The mass flow rate is first forced to be non-negative,

$$\dot{m} = \max(\dot{m}_{\text{PA}}, 0),$$

⁸Bell et al. 2014.

and the upstream stagnation enthalpy and entropy are computed from (P_1, T_{line}) :

$$h_1 = h(P_1, T_{\text{line}}), \quad s_1 = s(P_1, T_{\text{line}}).$$

In case of numerical problems with `CoolProp` (for instance near saturation), a fallback to saturated liquid ($Q = 0$) at pressure P_1 is used, still obtaining a physically consistent estimate of h_1 and s_1 .

2. **Isentropic downstream state.** The outlet is reconstructed by imposing the same entropy upstream and downstream, in analogy with the HEM model:

$$s_2 = s_1, \quad (P_2, s_2).$$

In practice, the code evaluates the thermodynamic state at $(P_2, s = s_1)$:

$$h_2 = h(P_2, s_1), \quad T_{\text{out}} = T(P_2, s_1).$$

If the isentropic call fails (for example for conditions very close to critical), an isothermal fallback is used:

$$T_{\text{out}} = T_{\text{line}}, \quad h_2 = h(P_2, T_{\text{out}}),$$

thus preserving a robust definition of the downstream state.

3. **Reconstruction of the mixture density (HEM).** The outlet density is obtained by interpreting the state (P_2, h_2) as a possible two-phase mixture by means of the `mixture_rho_HEM` function, which returns:

$$(\rho_{\text{mix}}, \rho_\ell, \rho_v, x, \text{is_two}),$$

where ρ_ℓ and ρ_v are the saturated liquid and vapour densities at P_2 , while x is the mass quality (mass fraction of vapour). If h_2 lies between the saturated-liquid and saturated-vapour enthalpies, a true two-phase mixture is obtained ($0 \leq x \leq 1$); otherwise, `mixture_rho_HEM` flags a single-phase regime. In case of single-phase regime or numerical issues, the code performs an explicit fallback based on the saturation temperature $T_{\text{sat}}(P_2)$:

- if $T_{\text{out}} > T_{\text{sat}}(P_2)$, a superheated gas is assumed and the density is evaluated via `rho_singlephase_at_T` on the gas branch;
- otherwise, subcooled liquid is assumed and the density is evaluated on the liquid branch.

In this way a well-defined downstream density ρ_2 is always obtained, with $x = 1$ (gas) or $x = 0$ (liquid) in single-phase conditions.

4. **Velocity, dimensionless numbers and void fraction.** Given the outlet density ρ_2 , the average outlet velocity is directly obtained from the mass flow rate:

$$A = \frac{\pi D^2}{4}, \quad U_{\text{out}} = \frac{\dot{m}}{\rho_2 A}.$$

The effective mixture viscosity is calculated through the `_safe_viscosity` function:

- in two-phase conditions ($0 < x < 1$) an HEM-type blend is used:

$$\mu_{\text{mix}} = (1 - x) \mu_\ell + x \mu_g,$$

with μ_ℓ and μ_g evaluated on the liquid and gas branches, respectively;

- in single-phase regime the viscosity of the branch consistent with x (gas or liquid) is used, with small pressure adjustments to avoid instabilities near saturation.

The speed of sound is reconstructed by `_safe_speed_of_sound`: in two-phase regime it is approximated with the sonic speed on the gas branch at $T_{\text{sat}}(P_2)$, while in single-phase regime the standard real-fluid relation $a(P, T)$ is used. The following are then obtained:

$$M = \frac{U_{\text{out}}}{a_{\text{out}}}, \quad Re_{\text{out}} = \frac{\rho_2 U_{\text{out}} D}{\mu_{\text{mix}}}.$$

The outlet void fraction α_{out} is calculated from the quality x and the phase densities:

$$V_v = \frac{x}{\rho_v}, \quad V_\ell = \frac{1 - x}{\rho_\ell}, \quad \alpha_{\text{out}} = \frac{V_v}{V_v + V_\ell},$$

with fallback to $\alpha_{\text{out}} \approx 0$ (liquid) or $\alpha_{\text{out}} \approx 1$ (gas) in single-phase cases.

5. **Volumetric flow rates and superficial velocities.** From the total mass flow rate one obtains the total volumetric flow rate

$$\dot{V} = \frac{\dot{m}}{\rho_2},$$

which is split into liquid and gas contributions as a function of x :

$$\dot{V}_v \approx \frac{x \dot{m}}{\rho_v}, \quad \dot{V}_\ell \approx \frac{(1 - x) \dot{m}}{\rho_\ell},$$

with appropriate fallbacks in single-phase regime. The superficial velocities are then:

$$j_\ell = \frac{\dot{V}_\ell}{A}, \quad j_g = \frac{\dot{V}_v}{A}.$$

The `nhne_out_state_from_mdot` function returns a dictionary containing all these quantities (`T_out`, `rho_mix`, `mu_mix`, `a_out`, `U_out`, `Mach`, `Re_out`, `x_out`, `alpha_out`, `j_liq`, `j_gas`, phase densities, phase label "liquid"/"gas"/"two-phase", etc.), which is used by `PyInjection_performance` to populate the tables and by `PyInjection_design` to characterise the flow regime associated with each candidate geometry. In this way the reconstruction of the outlet state is fully consistent with the isentropic-expansion assumption adopted for the HEM mass-flow model.

2.2.4 Semi-empirical model of the discharge coefficient C_d

The estimation of the discharge coefficient is handled by the high-level function `estimate_Cd_from_geometry`, which provides a single interface for two operating modes: *use of an imposed C_d* or *estimation of C_d from geometry*.

Mode with imposed C_d

If the user specifies a value `Cd_input` > 0, the code uses it directly:

$$C_d = C_{d,\text{input}}.$$

In this case, the function only calculates a diagnostic Reynolds number, through `_Re_from_mdot_target`:

$$Re_{\text{char}} = \frac{4 \dot{m}_{\text{target}}}{\pi \mu_{\text{out}} D_{\text{orif}}}.$$

The properties at the upstream state `rho_out` and `mu_out` are nevertheless evaluated with `CoolProp`:

$$\rho_{\text{out}} \approx \rho(P_1, T_{\text{line}}), \quad \mu_{\text{out}} \approx \mu(P_1, T_{\text{line}}, \text{liq}).$$

Geometry-based estimation mode.

If `Cd_input` is not provided, `estimate_Cd_from_geometry` calls the internal function `estimate_Cd_geom_full`, which implements a semi-empirical model for short orifices, built from:

- ISO 5167 Reader-Harris/Gallagher correlation for sharp-edged orifices;⁹
- local and distributed loss-coefficient recalls from process-engineering handbooks and from studies on liquid injectors;¹⁰
- experimental data and simulations on short orifices and cavitating injectors, with explicit dependence on Re , L/D and r/D .¹¹

⁹ISO 5167-2; Reader-Harris and Sattary 1998.

¹⁰Perry and Green 1984; Sutton 1976.

¹¹Metwally 2014; Sun et al. 2015; Lee et al. 2018; Vargas Niño and Razavi 2019; Zimmerman et al. 2012; Waxman et al. 2019.

The logic implemented in `estimate_Cd_geom_full` can be summarised as follows.

1. Reynolds number at the orifice

Given a target mass flow rate $\dot{m}_{\text{target}}$, the characteristic Reynolds number is:

$$Re_{\text{or}} = \frac{4 \dot{m}_{\text{target}}}{\pi \mu_{\text{out}} D_{\text{orif}}},$$

with D_{orif} the orifice diameter and μ_{out} the viscosity evaluated on the liquid branch.

For the ISO 5167 correlation one also defines:

$$\beta = \frac{D_{\text{orif}}}{D_{\text{pipe}}}, \quad Re_D = \beta Re_{\text{or}}.$$

The dependence of the discharge coefficient on the turbulent regime is consistent with what is observed in studies on two-phase injectors and on short orifices.¹²

2. Base coefficient for sharp-edged orifice

The `_Cd_reader_harris_gallagher` function explicitly implements the Reader – Harris / Gallagher (RHG) correlation of ISO 5167 for sharp-edged orifice plates.¹³ The correlation was obtained from a database of about 1.1×10^4 experimental measurements for a water–air mixture in turbulent regime, in the range $0.1 \lesssim \beta \lesssim 0.75$ and $4 \times 10^3 \lesssim Re_D \lesssim 10^7$.

In the code, after limiting the validity field, one first defines:

$$A = \left(\frac{19000 \beta}{Re_D} \right)^{0.8}, \quad M'_2 = \frac{2L'_2}{1 - \beta},$$

where L_1 and L'_2 depend on the type of pressure taps (`tap_type`). For "corner" taps (case adopted in this work) one has $L_1 = 0$ and $L'_2 = 0$.

The analytical form of the base discharge coefficient is:

$$\begin{aligned} C_{d,\text{RHG}} = & 0.5961 + 0.0261 \beta^2 - 0.216 \beta^8 + 0.000521 \left(\frac{10^6 \beta}{Re_D} \right)^{0.7} \\ & + (0.0188 + 0.0063 A) \beta^{3.5} \left(\frac{10^6}{Re_D} \right)^{0.3} \\ & + \left(0.043 + 0.080 e^{-10L_1} - 0.123 e^{-7L_1} \right) (1 - 0.11 A) \frac{\beta^4}{1 - \beta^4} \\ & - 0.031 (M'_2 - 0.8 M'^{1.1}_2) \beta^{1.3}. \end{aligned}$$

¹²Vargas Niño and Razavi 2019; Zimmerman et al. 2012; Waxman et al. 2019.

¹³ISO 5167-2; Reader-Harris and Sattary 1998.

For small pipe diameters ($D_{\text{pipe}} < 71.2$ mm) the ISO correction is applied in the code:

$$C_{d,\text{RHG}} \leftarrow C_{d,\text{RHG}} + 0.011 (0.75 - \beta) \left(2.8 - \frac{D_{\text{pipe}}}{0.0254} \right),$$

in order to maintain consistency with the original experimental dataset.

In compact form, the dependence can thus be summarised as:

$$C_{d,\text{RHG}} = C_{d,\text{RHG}}(\beta, Re_D, D_{\text{pipe}}, \text{tap_type}),$$

with `tap_type` = "corner" in the present case. This correlation represents the “sharp-edged thin plate” limit of the `estimate_Cd_geom_full` model.

3. Gain due to inlet rounding r/D

The inlet rounding is modelled by:

$$G_r(r/D) = 1 + \text{gain_max} \left[1 - \exp\left(-\frac{r/D}{r_{\text{sat}}}\right) \right],$$

with default values `gain_max` = 0.25, `r_sat` = 0.08.

The corrected base coefficient is:

$$C_{d,\text{eq}} = C_{d,\text{RHG}} G_r(r/D).$$

For $r/D \gtrsim 0.2$ increases of C_d of 20–25% are observed, in agreement with experimental studies on rounded-inlet orifices and nozzles.¹⁴

4. Distributed losses in the cylindrical section (L/D)

The internal viscous loss is modelled by the Darcy–Weisbach relation:

$$K_D = f(Re_{\text{or}}, \varepsilon/D_{\text{orif}}) \frac{L}{D_{\text{orif}}},$$

where the friction factor is:

$$f = \begin{cases} \frac{64}{Re}, & Re < 2300, \\ \frac{0.25}{\left[\log_{10} \left(\frac{\varepsilon/D}{3.7} + \frac{5.74}{Re^{0.9}} \right) \right]^2}, & Re \geq 2300. \end{cases}$$

The equivalent roughness is set to `eps_abs` = 2×10^{-6} m, in line with machined surfaces.¹⁵ The combined dependence on L/D and flow regime is consistent with the literature on short orifices and N₂O injectors.¹⁶

¹⁴Metwally 2014; Lee et al. 2018; Sun et al. 2015; Vargas Niño and Razavi 2019.

¹⁵Perry and Green 1984.

¹⁶Metwally 2014; Vargas Niño and Razavi 2019; Zimmerman et al. 2012; Waxman et al. 2019.

5. Final combination in terms of C_d

The shape contribution (sharp edge + rounding) is:

$$K_{\text{shape}} = \frac{1}{C_{d,\text{eq}}^2} - 1.$$

The total loss is:

$$K_{\text{tot}} = K_{\text{shape}} + K_D.$$

The equivalent discharge coefficient is therefore:

$$C_d = \frac{1}{\sqrt{1 + K_{\text{tot}}}}.$$

For numerical robustness a final constraint is applied:

$$0.30 \leq C_d \leq 0.95.$$

In summary, the model implemented in `estimate_Cd_geom_full`:

- recovers the ISO 5167 correlation in the sharp-edged orifice limit;¹⁷
- incorporates the experimentally documented effects of Re , L/D and r/D for short orifices and injectors of self-pressurising oxidisers;¹⁸
- uses standard formulations of distributed and local losses taken from reference handbooks.¹⁹

It therefore represents a reliable preliminary estimate of C_d in the Reynolds and geometric range typical of the N_2O injectors analysed in this work.

2.3 `pyinjection_performance` module

The `pyinjection_performance.py` module provides a controlled environment for the detailed analysis of a single case of plain-orifice injector fed with N_2O .

This file does not introduce new physics, but simply calls the core functions, manages the inputs and organises the outputs in a readable form, both on the console and through a simple graphical interface.

¹⁷ISO 5167-2; Reader-Harris and Sattary 1998.

¹⁸Sutton 1976; Metwally 2014; Sun et al. 2015; Lee et al. 2018; Vargas Niño and Razavi 2019; Zimmerman et al. 2012; Waxman et al. 2019.

¹⁹Perry and Green 1984.

2.3.1 General structure and usage modes

The module can be used in two main modes:

- **Command-line mode.** Inputs are read through `argparse` (pressures, line temperature, orifice geometry, discharge coefficient, etc.). A typical command is, for example:

```
python pyinjection_performance.py --p1 55 --p2 43 --Tline 288
--D 2e-3 --L 10e-3 --Cd 0.85 --Cd-mode user
```

- **Graphical mode (GUI).** If the file is launched without arguments, the `-gui` functionality is automatically activated and a Tkinter window opens for interactive parameter input and case execution.

In both cases, the “engine” is the high-level function `run_performance_case`, which executes the whole numerical workflow and calls the graphical routine `show_output_tables_gui` for the visualisation of the results.

2.3.2 Computation flow

Given a set of input parameters, the internal workflow is as follows.

1. Determination of the discharge coefficient.

Depending on the chosen mode:

- *user mode*: the value of C_d provided by the user is directly used in all mass-flow models;
- *geom mode*: a local wrapper (`estimate_Cd_from_geometry_perf`) calls the `estimate_Cd_from_geometry` function from the core, using as input the single-hole geometry (D , L , r), the upstream manifold diameter D_{pipe} and a reference ideal mass flow rate. The function returns a value of C_d and an associated characteristic Reynolds number.

2. Calculation of single-hole and total mass flow rates.

With the value of C_d determined in the previous step, the `run_performance_case` function calculates the single-hole mass flow rates with the three models:

$$\dot{m}_{\text{SPI}}, \quad \dot{m}_{\text{HEM}}, \quad \dot{m}_{\text{NHNE}},$$

and the *phase-aware* mass flow rate via `compute_mdot_phaseaware`. Total mass flow rates are then obtained by multiplying by the number of holes N_h ; in the output tables, the reported values are always *total*, while the thermodynamic properties refer to a single hole.

3. Reconstruction of the outlet state.

Once the two-phase/metastable mass flow rate has been determined, the `nhne_out_state_from_mdot` function is used to build a consistent thermodynamic state at the outlet: temperature, mixture density and viscosity, vapour quality, void fraction, average velocity, Reynolds and Mach numbers, speed of sound, phase properties (liquid and gas) and superficial velocities.

4. Organisation and visualisation of the results.

The results are passed to the `show_output_tables_gui` function, which builds a Matplotlib window containing four tables:

- **input** table (geometric and thermodynamic parameters, and model options);
- **mass-flow** table (SPI, HEM, NHNE, phase-aware and selected model);
- **phase properties** table at the outlet (liquid and vapour, or single phase);
- **main mixture results** table (total mass flow rate, prevailing phase, T_{out} , $T_{\text{sat}}(P_2)$, U , ρ , μ , Re , M , surface tension).

This structure makes it possible to use the module as a “bridge” between the 0D model and CFD simulations: for each considered case, one has both a complete diagnostics of the mass-flow models (SPI/HEM/NHNE/phase-aware) and a block of thermodynamic properties directly comparable with numerical results.

2.3.3 Graphical input interface

The `launch_gui` function creates a Tkinter window that allows one to set all the case parameters and run the calculation via the *Run performance* button. The GUI is organised as follows:

- **Thermodynamic parameters:** fluid (string compatible with CoolProp), line temperature T_{line} , pressures P_1 and P_2 ;
- **Injector geometry:** per-hole diameter D , length L , rounding radius r , number of holes N_h and equivalent diameter D_{ref} ; the *Keep total area* option makes it possible to keep the total area A_{tot} constant when changing N_h , automatically varying D ;
- **C_d settings:** numerical value of C_d , mode `user` / `geom`, upstream manifold diameter D_{pipe} and information row *Cd source* describing the origin of the coefficient actually used;
- **SPI options:** activation/deactivation of the compressibility correction and value of the polytropic exponent n ;

- **Derived quantities:** read-only fields reporting L/D , r/D , per-hole area A_h and total area A_{tot} .

Each main field is equipped with informative *tooltips* explaining the physical meaning of the variable and its role in the model. A possible GUI screenshot is shown in Fig. 2.1.

Parameters	
Fluid	NitrousOxide
T_line [K]	288.0
P1 [bar]	55.0
P2 [bar]	43.0
D [m] (per hole)	0.002000
D_ref [m] (equiv. single)	0.002000
D_pipe [m]	-
L [m]	-
L/D [-]	-
r [m]	-
r/D [-]	-
Nh [-]	1
A_h [m^2]	3.14159265e-06
A_tot [m^2]	3.14159265e-06
Cd [-]	1.0
n SPI (opt.)	1.2
Cd mode	user
Cd source	user-provided Cd
☐ Keep total area A_tot constant when Nh changes	
☒ SPI compressible	
Run performance	

Figure 2.1: Example of graphical input interface.

2.3.4 Output window and result tables

At the end of the calculation, the `show_output_tables_gui` function generates a Matplotlib window that summarises the case in four tables:

- **Input parameters** (top left): summary of all input parameters, including L/D , r/D , per-hole area, total area, C_d mode and model options;
- **Mass flow table** (bottom left): total mass flow rates predicted by the three SPI/HEM/NHNE models and by the phase-aware solution, with indication of the selected model;
- **Outlet properties** (top right): phase properties at the outlet (temperature, density, viscosity, Reynolds number, volume fraction, heat capacity, enthalpy);
- **Main results** (bottom right): mixture properties (total mass flow rate, prevailing phase, T_{out} and $T_{\text{sat}}(P_2)$, average velocity, ρ , μ , Re , Mach number, surface tension).

An example of the output window is shown in Fig. 2.2.

INPUT PARAMETERS		OUTLET PROPERTIES (per hole)		
Parameter	Value	Phase	liquid	vapor
Fluid	NitrousOxide	T [K]	286.176	286.176
T_line [K]	293.000	T_sat [K]	286.176	286.176
P1 [bar]	55.000	U [m/s]	43.668	22.710
P2 [bar]	43.000	rho [kg/m3]	833.540	126.132
Orifice D_hole [m]	2.000e-03	mu [Pa.s]	3.000e-04	1.850e-05
D_ref [m] (equiv.)	2.000e-03	Re [-]	2.427*10 ⁻⁵	3.097*10 ⁻⁵
Orifice L [m]	-	alpha [-]	0.658	0.342
L/D [-]	-	Cp [J/kgK]	2703.914	2328.915
Edge radius r [m]	-	MW [kg/kmol]	44.013	44.013
r/D [-]	-	h [J/kg]	1.960e+05	3.913e+05
Nh [-]	1			
A_h [m ²]	3.142e-06			
A_tot [m ²]	3.142e-06			
Cd [-]	1.000			
Cd mode	user			
Cd source	user-provided Cd			
Keep A_tot const (Nh)	OFF			
SPI compressible	OFF			
n SPI	1.200			

MASS FLOW TABLE - TOTAL mdot (SPI / HEM / NHNE / Phase-aware)		MAIN RESULTS - TOTAL mass flow and mixture properties	
Model	mdot_total [kg/s]	Parameter	Value
SPI	0.137	mdot_total [kg/s]	0.123
HEM	0.106	Phase	two-phase
NHNE	0.123	T [K]	286.176
Phase-aware	0.123	T_sat [K]	286.176
		U_mix [m/s]	66.378
		rho_mix [kg/m3]	591.513
		mu_mix [Pa.s]	2.795e-04
		Re_mix [-]	2.810*10 ⁻⁵
		Mach [-]	0.325
		sigma [N/m]	0.003

Figure 2.2: Example of graphical output interface.

2.4 pyinjection_design module

The `pyinjection_design.py` module implements the *inverse-design* tool for the plain-orifice. Unlike the `pyinjection_performance` module, which analyses a single geometry at a time, here the focus is on a systematic exploration of the geometric space to identify the solutions that satisfy a mass-flow requirement imposed by the propulsion system.

2.4.1 Search on a three-dimensional grid

The central function is `design_from_mdot`. Given fluid, pressures, line temperature, target mass flow rate $\dot{m}_{\text{target}}$ and number of holes N_h , the user specifies the ranges and discretisations of the three geometric parameters:

$$r/D \in [rD_{\min}, rD_{\max}], \quad D \in [D_{\min}, D_{\max}], \quad L/D \in [(L/D)_{\min}, (L/D)_{\max}].$$

In this project the upstream pipe diameter D_{pipe} is fixed a priori and supplied as an input parameter to the code. Consequently, the design procedure explores only the range of orifice diameters D and of the ratios r/D and L/D , without treating the dimensionless ratio D/D_{pipe} as a design variable. In future versions of the tool the grid may be extended to include D/D_{pipe} among the parameters to be optimised, so as to explicitly evaluate the influence of the ratio between orifice diameter and line diameter on the discharge coefficient and on distributed losses.

The function generates a regular grid in the space $(r/D, D, L/D)$ and, for each point:

1. computes the length $L = (L/D) D$;
2. estimates the discharge coefficient C_d by calling `estimate_Cd_from_geometry`, except in the case where the user imposes a fixed value (for example derived from CFD);
3. computes the single-hole mass flow rate via the phase-aware backend `compute_mdot_phaseaware`;
4. reconstructs the downstream state with `nhne_out_state_from_mdot` and derives from it the mixture density and viscosity, used to compute the effective Reynolds number;
5. determines the total mass flow rate by multiplying by the number of holes N_h ;
6. stores all the information in a `CandidateResult` structure, which contains geometry, C_d , Re, per-hole mass flow rate, total mass flow rate and some reference flow properties.

The execution on the grid can take place in serial mode or in parallel, using a `ProcessPoolExecutor` to speed up the calculation when the number of combinations is large.

2.4.2 Candidate selection and final Pareto-plot

Once the list of candidates has been populated, the module applies a set of selection criteria:

- a relative-error function

$$\varepsilon = \frac{|\dot{m}_{\text{tot}} - \dot{m}_{\text{target}}|}{\dot{m}_{\text{target}}}$$

allows all configurations to be sorted based on the distance from the required mass flow rate;

- `select_best_candidates` returns the candidates with minimum relative error (for example the best ten), useful for detailed numerical analysis;
- `get_feasible_candidates` filters the set of solutions by applying a tolerance threshold

$$\varepsilon \leq \varepsilon_{\text{max}},$$

expressed as a percentage.

The only plot used in the project is generated by the `plot_cd_vs_ratio_by_diameter` function. The plot shows, for all configurations that satisfy the mass-flow constraint within the chosen tolerance:

- on the horizontal axis the discharge coefficient C_d ;
- on the vertical axis the ratio $\dot{m}_{\text{tot}}/\dot{m}_{\text{target}}$;
- the colour of the points used to distinguish the different diameters D ;
- a horizontal reference line at $\dot{m}_{\text{tot}}/\dot{m}_{\text{target}} = 1$.

Each point in the plot is associated with a **CandidateResult** object and can display, through an interactive tooltip, the values of L/D , r/D , Re and C_d related to that geometry. The diagram can be interpreted as a Pareto front between efficiency (high values of C_d) and adherence to the target mass flow (ratio close to 1). In the text of the report it will be discussed how, for each diameter, there generally exists an interval of admissible geometries with variable C_d , and how the final choice must represent a compromise between maximisation of the discharge coefficient and consistency with the target mass-flow requirement.

In addition, the ranges assigned to the geometric ratios r/D and L/D must be defined consistently with the actual constraints of the component. In particular, the lower and upper limits must be chosen taking into account the real manufacturing possibilities, machining tolerances, structural requirements, thermal loads in operation and integration constraints of the injector within the propulsion system.

In the tool interface, typical default values for r/D and L/D are also proposed, derived from the literature on short orifices and from common manufacturing practices; these values provide an initial reference for the user, who can nevertheless adapt the ranges according to the design specifications of their case study.

2.4.3 Graphical interface

The module finally includes a simple graphical interface (**run_gui**) based on Tkinter. The GUI makes it possible to:

- interactively enter the operating conditions (pressures, temperature) and the ranges of the design parameters (r/D , D , L/D , number of holes, relative-error tolerance, possible imposed C_d , D_{pipe});
- start the grid calculation and display on the console a table of the best candidates;
- generate the Pareto-style plot described above.

As an example, Fig. 2.3 shows the main GUI screen of the **pyinjection_design** module, in which the operating parameters and the geometric ranges to be explored are specified.

Fluid (CoolProp):	NitrousOxide
P1 [bar]:	55.0
P2 [bar]:	43.0
T_line [K]:	288.0
mdot target total [kg/s]:	0.140
Nh (number of holes):	1
r/D min:	0.05
r/D max:	0.35
n_rD:	7
D_min [mm]:	0.5
D_max [mm]:	3.5
nD:	25
D_pipe [mm]:	5.0
L/D min:	2.0
L/D max:	12.0
n_LD:	10
tol. rel. [%]:	3.0
n_workers:	6
Cd fixed (blank = auto):	
n SPI (optional):	1.2
☒ Use SPI compressible correction	
Run design	

Figure 2.3: Example of graphical input interface.

Once the scan of the three-dimensional grid has been completed, the GUI returns a table containing the best candidates in terms of relative error on the mass flow rate. The most significant output of the module is the Pareto plot generated by the `plot_cd_vs_ratio_by_diameter` function. An example is shown in Fig. 2.4. The diagram shows, for all geometries that satisfy the target mass flow within the specified tolerance:

- on the horizontal axis the discharge coefficient C_d ;
- on the vertical axis the ratio $\dot{m}_{\text{tot}}/\dot{m}_{\text{target}}$;
- colour indicates the orifice diameter D ;
- a horizontal line highlights the ideal solution ($\dot{m}_{\text{tot}}/\dot{m}_{\text{target}} = 1$).

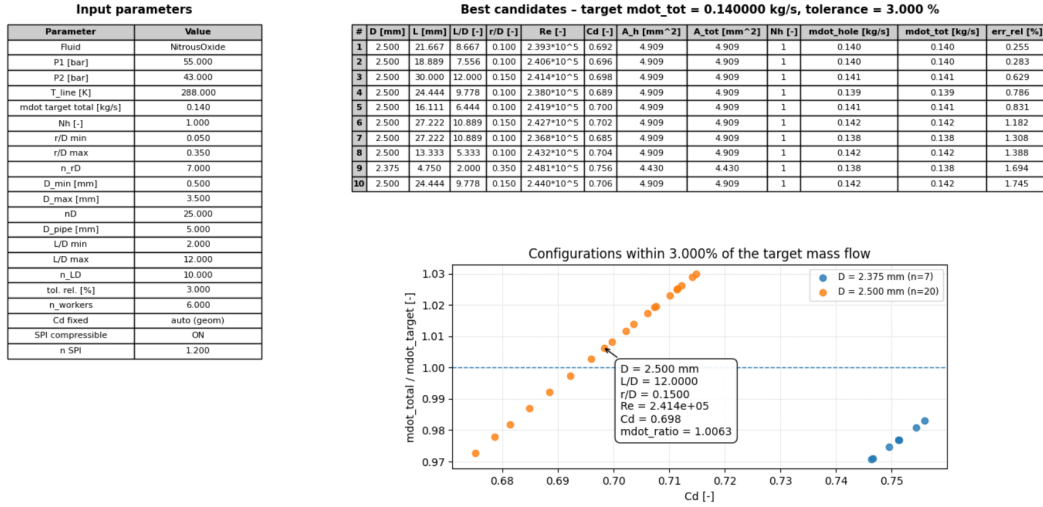


Figure 2.4: Example of Pareto front generated by *pyinjection_design*.

In the Pareto plot, each point is linked to a **CandidateResult** object. By hovering the cursor over a point, the GUI shows through a tooltip:

- the exact geometric values of the configuration (r/D , D , L/D);
- the discharge coefficient C_d predicted by the model;
- the effective Reynolds number;
- the normalised total mass flow rate.

This makes it possible to:

- quickly identify which regions of the (r/D , D , L/D) space produce high discharge coefficient;
- understand how the regime varies along the Pareto front;
- consciously select the optimal compromise between hydraulic efficiency (C_d) and accuracy with respect to the target mass flow rate.

The interactive cursor functionality turns the Pareto plot into not only a descriptive but also a diagnostic tool: the user can instantly trace back the geometric structure and physical model used for each solution on the frontier.

The module finally includes a simple graphical interface (**run_gui**) based on Tkinter. The GUI makes it possible to:

- interactively enter the operating conditions (pressures, temperature) and the ranges of the design parameters (r/D , D , L/D , number of holes, relative-error tolerance, possible imposed C_d , D_{pipe});

-
- start the grid calculation and display on the console a table of the best candidates;
 - generate the Pareto-style plot described above.

3. Validation with CFD

The validation of the analytical model developed in `PyInjection` was carried out by comparing the mass flow rates predicted by the SPI, HEM, NHNE models and by the *phase-aware* selector with CFD simulations performed in ANSYS Fluent. The simulations were set up in three regimes representative of the injector behaviour: a single-phase case, a weakly two-phase case, and a fully two-phase case obtained with a VOF-to-DPM methodology. This progression makes it possible to assess the model first in the liquid limit, then in conditions close to saturation and finally in the presence of significant vaporisation, nucleation and film formation.

The single-phase case provides the validation reference for the SPI model and for the isentropic assumption adopted in the computation of the downstream state in the HEM and NHNE models. The weakly two-phase configuration instead allows one to verify the sensitivity of the intermediate NHNE model and the effectiveness of the *phase-aware* selector in identifying the onset of a regime with limited two-phase relaxation. The fully two-phase case finally makes it possible to qualitatively compare the behaviour predicted by the model with a complete simulation of the phenomenon.

3.1 Validation of the SPI model

3.1.1 Case 1 – Nearly single-phase regime

This first validation test is performed under operating conditions such as to keep the N_2O flow almost entirely in the liquid state, with very weak two-phase effects. It therefore represents the ideal case for assessing the accuracy of the SPI model and verifying the consistency of the NHNE model in transition conditions towards the two-phase regime.

CFD simulation

The simulation was carried out in ANSYS Fluent in steady-state conditions, using a two-dimensional axisymmetric domain and a structured mesh. The boundary conditions imposed are:

- **Inlet (manifold):** Pressure Inlet with total pressure $P_0 = 55$ bar;

- **Outlet (chamber):** Pressure Outlet with static pressure $P = 43$ bar;
- **Walls:** *no-slip* condition.

The SST k - ω turbulence model was adopted, typically preferred for flows with moderate separation and well-resolved boundary layers. Convergence was monitored through the numerical residuals, kept below 10^{-6} for the momentum equations, as shown in Figure 3.2.

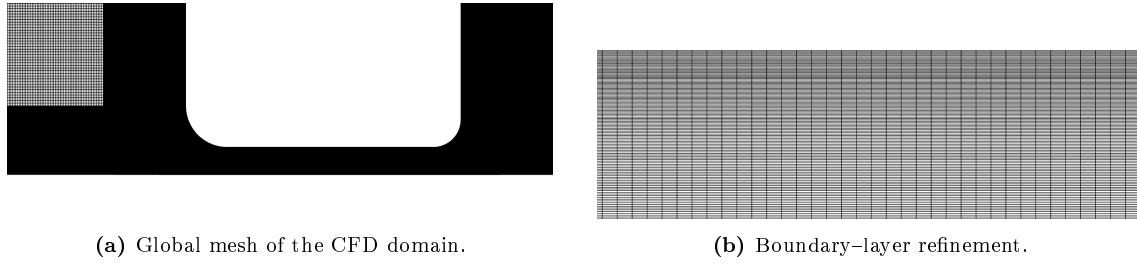


Figure 3.1: CFD mesh used.

After convergence, the actual pressures upstream and downstream of the orifice were extracted directly from the solution field, obtaining the velocity field shown in Figure 3.3.

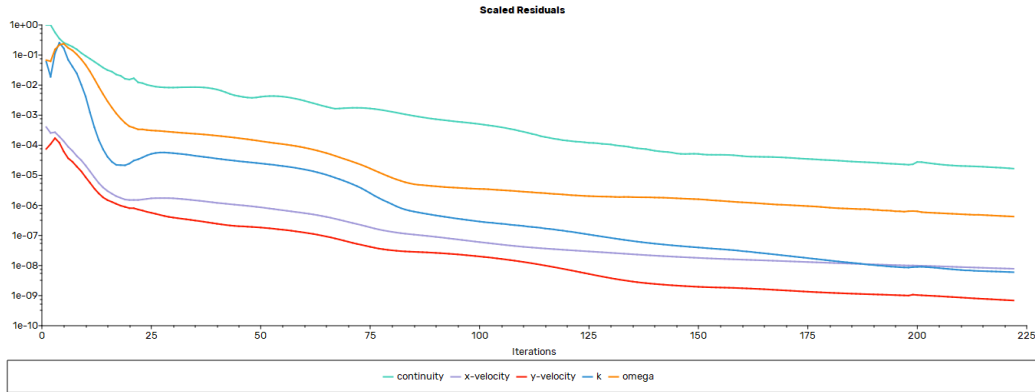


Figure 3.2: Evolution of numerical residuals.

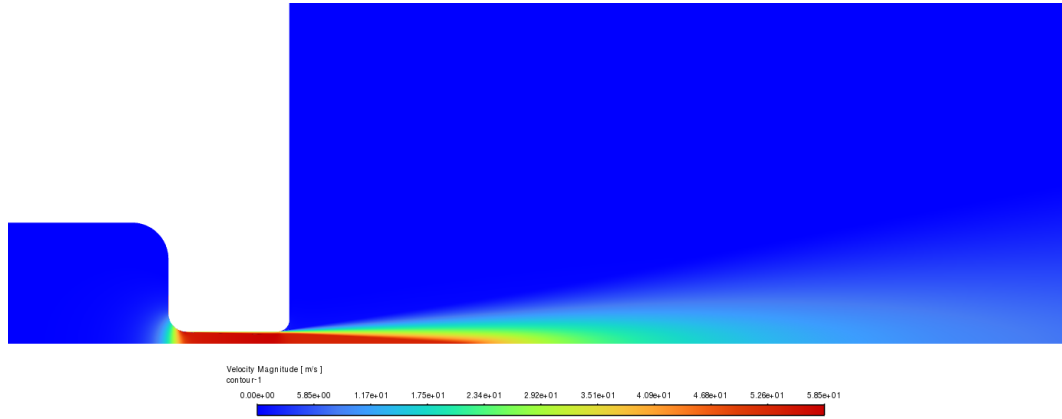


Figure 3.3: Velocity field in the orifice region.

The effective mass flow rate at the orifice outlet resulting from the simulation is:

$$\dot{m}_{\text{CFD}} = 0.138765 \text{ kg/s}, \quad C_{d,\text{CFD}} = 0.92725.$$

These values are obtained from the post-processing reported in Figure 3.4.

Mass Flow Rate	[kg/s]	Expression	[]
line-out	-0.13876508	cd	0.92725293

(a) Mass flow rate extracted from CFD. (b) Computation of the discharge coefficient in CFD.

Figure 3.4: CFD post-processing for mass flow rate and C_d (Case 1).

Important note: once the CFD solution was obtained, the actual pressures at the orifice inlet and outlet were measured directly from the solution field:

$$P_{1,\text{CFD}} = 54.4626 \text{ bar}, \quad P_{2,\text{CFD}} = 40.8510 \text{ bar},$$

values that were inserted into the analytical model to enable a consistent comparison.

Simulation with PyInjection

To consistently reproduce the conditions of the CFD simulation, the `PyInjection_performance` module was initialised using the pressure values measured at the orifice inlet and outlet. The input parameters are given in Table 3.1a, while the resulting mass flow rates (for the four implemented models) are summarised in Table 3.1b.

Parameter	Value		
Fluid	NitrousOxide		
T_{line} [K]	288		
P_1 [bar]	54.4626		
P_2 [bar]	40.8510		
Orifice diameter D [m]	0.002		
Orifice area [m ²]	3.1416×10^{-6}	Model	$\dot{m}$ [kg/s]
Number of holes N_h	1	SPI	0.139
C_d	0.92725 (from CFD)	HEM	0.119
SPI compressible	OFF	NHNE	0.132
n_{SPI}	1.2	Phase-aware	0.132

(a) Inputs of the analytical model. (b) Mass flow rates predicted by the analytical models.

Table 3.1: Summary of inputs and predicted mass flow rates for the nearly single-phase case.

Finally, for completeness, Figure 3.5 shows all the inputs and outputs related to the performed simulation.

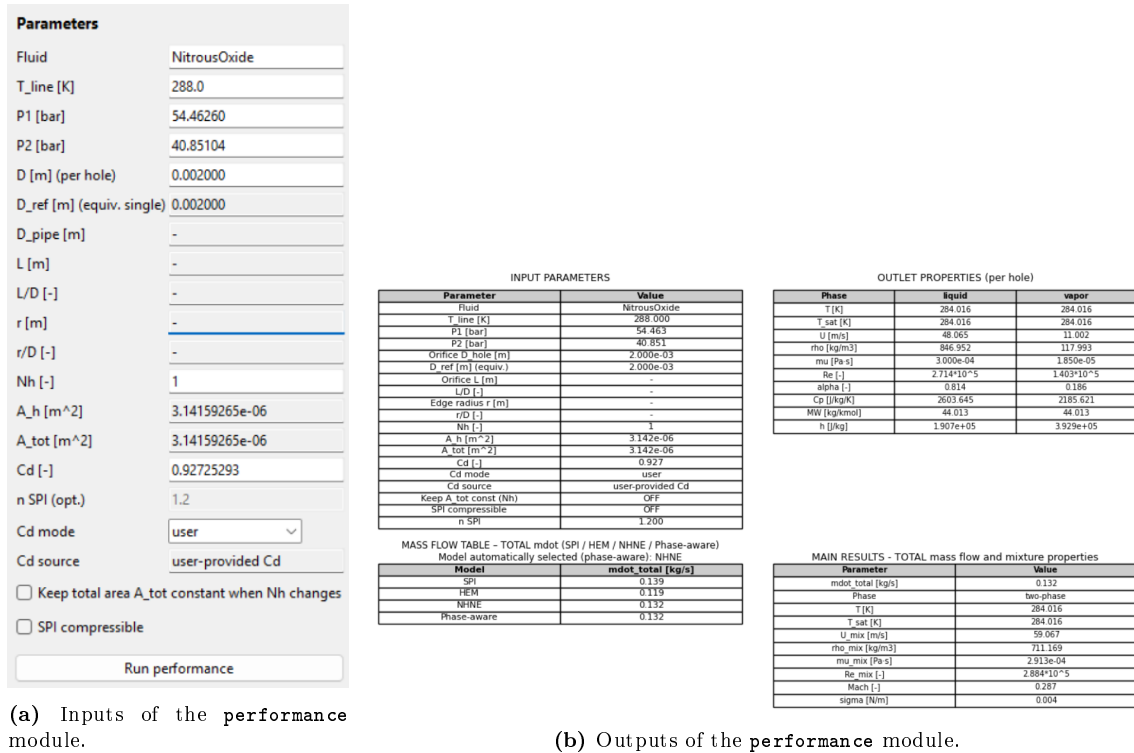


Figure 3.5: Interface of the performance module.

Comparison and discussion

The quantitative comparison between the SPI model and CFD is reported in Table 3.2.

Table 3.2: Comparison between SPI model and CFD (Case 1).

Model	$\dot{m}$ [kg/s]	Error with respect to CFD
SPI	0.139	+0.18%
CFD	0.138765	–

From this comparison it emerges that the relative error of 0.18% confirms that, in nearly single-phase conditions, the SPI model reproduces the CFD mass flow rate with excellent accuracy. SPI indeed represents the correct physical limit when no significant two-phase relaxation occurs.

Overall, the nearly single-phase case confirms that

$$\dot{m}_{\text{SPI}} \approx \dot{m}_{\text{CFD}},$$

thus validating the single-phase model as an upper reference limit for the mass flow rate, to be used as a baseline for the analysis of the two-phase models in the following sections.

3.1.2 Case 2 – More pronounced two-phase regime

In this second test, an operating configuration with a higher pressure drop than Case 1 is analysed, such as to enhance vapour formation while keeping the flow globally dominated by the liquid phase. The case is useful to investigate the behaviour of the two-phase models (HEM and NHNE) and of the *phase-aware* selector in the presence of a more pronounced two-phase regime.

CFD simulation

The simulation was carried out in ANSYS Fluent in steady-state conditions on a two-dimensional axisymmetric domain, reusing the same structured mesh as in Case 1 (Figure 3.1), with refinement in the orifice region and in the boundary layer. The imposed boundary conditions are:

- **Inlet (manifold):** Pressure Inlet with total pressure $P_0 = 65$ bar;
- **Outlet (chamber):** Pressure Outlet with static pressure $P = 43$ bar;
- **Walls:** *no-slip* condition.

Also in this case, the SST $k-\omega$ turbulence model was adopted. Convergence was monitored through the numerical residuals, kept below 10^{-6} for the momentum equations (Figure 3.6), while the velocity field in the orifice region is shown in Figure 3.7.

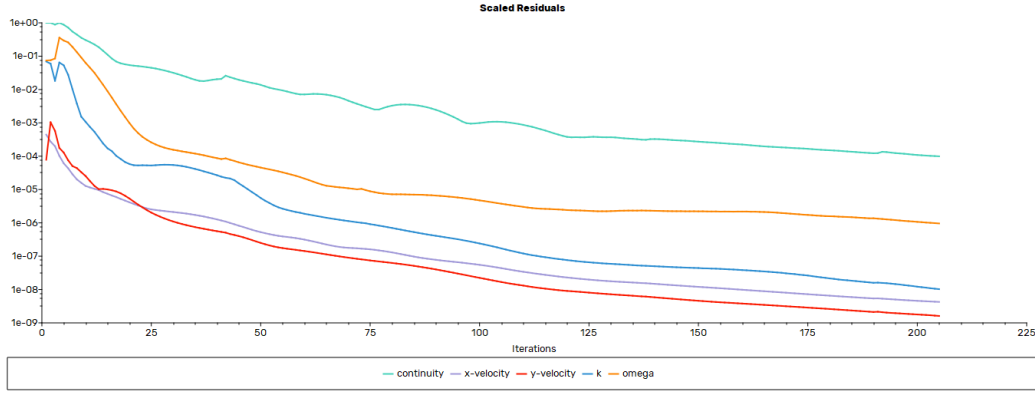


Figure 3.6: Evolution of numerical residuals (Case 2).

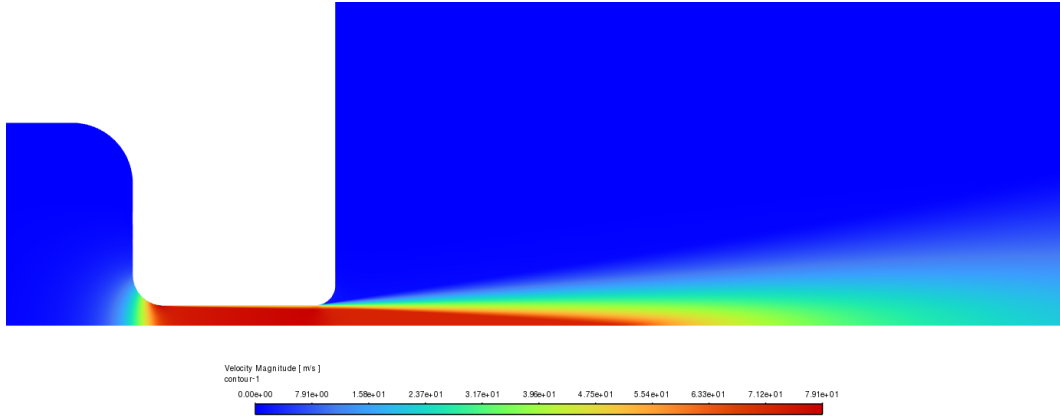


Figure 3.7: Velocity field in the orifice region (Case 2).

Once convergence was reached, the actual pressures upstream and downstream of the orifice were measured directly from the solution field:

$$P_{1,\text{CFD}} = 64.1079 \text{ bar}, \quad P_{2,\text{CFD}} = 39.0063 \text{ bar}.$$

The mass flow rate at the outlet and the corresponding discharge coefficient were obtained through appropriate *report definitions*; the results are summarised in Figure 3.8:

$$\dot{m}_{\text{CFD}} = 0.18835 \text{ kg/s}, \quad C_{d,\text{CFD}} = 0.92522.$$

Mass Flow Rate	[kg/s]	Expression	[]
line-p2	-0.18834763	cd	0.92522179

(a) Mass flow rate extracted from CFD.

(b) Computation of the discharge coefficient C_d in CFD.

Figure 3.8: CFD post-processing for mass flow rate and C_d (Case 2).

As in Case 1, the values $P_{1,\text{CFD}}$, $P_{2,\text{CFD}}$ and $C_{d,\text{CFD}}$ were used as inputs in the 0D model to guarantee a consistent comparison.

Simulation with PyInjection

The `PyInjection_performance` module was initialised using the pressures measured in CFD and the value of C_d obtained from the simulation. The main inputs and the mass flow rates predicted by the four analytical models are summarised in Table 3.3.

Parameter	Value		
Fluid	NitrousOxide		
T_{line} [K]	288		
P_1 [bar]	64.1079		
P_2 [bar]	39.0063		
Orifice diameter D [m]	0.002		
Orifice area [m ²]	3.1416×10^{-6}	Model	$\dot{m}$ [kg/s]
Number of holes N_h	1	SPI	0.190
C_d	0.92522 (from CFD)	HEM	0.154
SPI compressible	OFF	NHNE	0.178
n_{SPI}	1.2	Phase-aware	0.178

(a) Inputs of the analytical model.

(b) Mass flow rates predicted by the analytical models.

Table 3.3: Summary of inputs and predicted mass flow rates for Case 2.

For completeness, Figure 3.9 shows the graphical interface of the `performance` module with all the inputs and outputs related to this case.

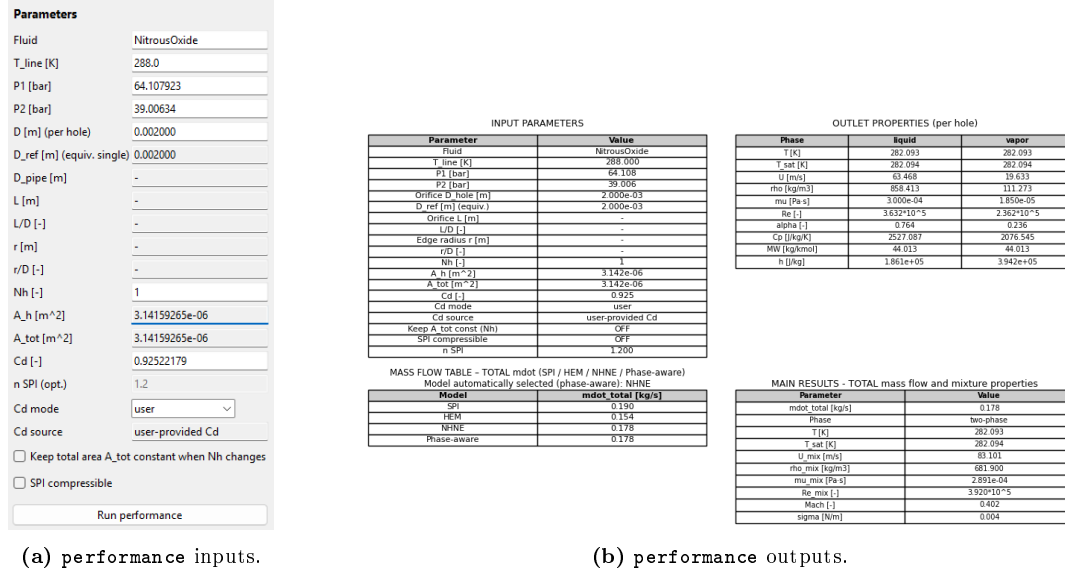


Figure 3.9: Interface of the performance module for Case 2.

Comparison and discussion

Table 3.4 reports the quantitative comparison between the SPI model and the value obtained from the CFD simulation.

Table 3.4: Comparison between SPI model and CFD (Case 2).

Model	$\dot{m}$ [kg/s]	Error with respect to CFD
SPI	0.190	+0.9%
CFD	0.18835	—

Also in this case the SPI model proves to be very accurate, with an error lower than 1%, and continues to represent the reference single-phase limit. The increased pressure drop and the closer proximity to saturation, however, enhance the potential two-phase component of the flow, but the comparison shows that under these operating conditions the flow remains globally dominated by the liquid phase and the single-phase model is still able to correctly reproduce the total mass flow rate. Overall, Case 2 confirms the robustness of the single-phase SPI limit in operating conditions still dominated by the liquid phase, and provides a second validation point for the use of the model as a reference in the subsequent analysis of the two-phase models.

It should be noted that the Fluent simulations of the first two cases were carried out in single-phase liquid regime. Therefore, the comparison with the analytical models is fully meaningful only for the single-phase limit represented by SPI. The differences observed with NHNE, HEM and with the *phase-aware* selector reflect

the physical structure of the two-phase models but do not constitute a direct measure of the two-phase accuracy, which will be assessed in the fully two-phase case through dedicated simulations.

3.2 Validation of the NHNE model

The validation of the two-phase NHNE model and of the *phase-aware* selector was carried out by means of a fully two-phase CFD simulation, developed with a VOF-to-DPM methodology on a three-dimensional domain including both the feed manifold and a portion of chamber downstream of the orifice. The main objective is to verify the consistency of the NHNE model in the flashing regime, by comparing the mass flow rate predicted by the 0D model with the CFD simulation and deriving an effective discharge coefficient C_d in two-phase conditions.

The simulation was set up in incompressible regime, with the energy equation disabled and thermophysical properties depending on a reference temperature, so as to isolate the purely hydraulic contribution of the two-phase relaxation. The evaporation of N_2O is not explicitly modelled at this stage: vapour formation is represented by the VOF interface dynamics and the mass transfer towards the discrete DPM phase, while the introduction of a complete thermal model with real-gas and evaporation effects is left to future developments.

3.2.1 Numerical setup and physical model

The CFD domain reproduces the manifold already sized for the propulsion system, the upstream line section, the plain-orifice and a sufficiently long downstream expansion chamber. The initial mesh, of hybrid hex-tet type with local refinement in the orifice region, is shown in Figure 3.10.

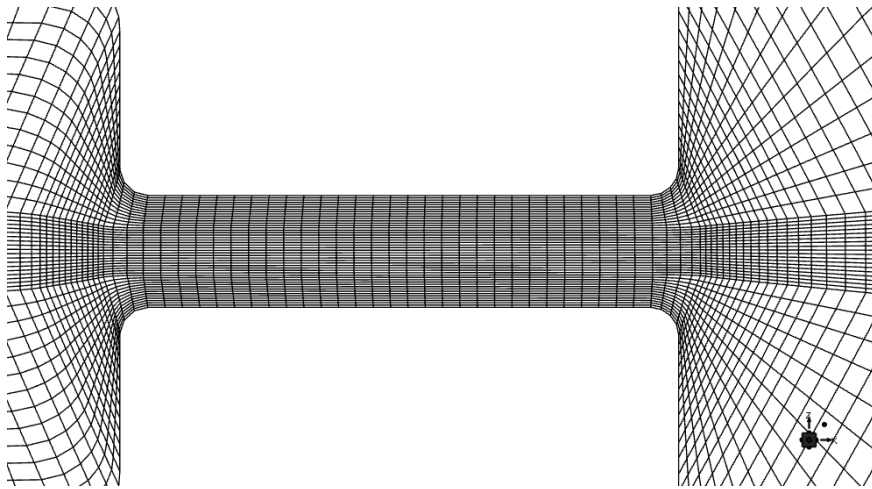


Figure 3.10: Initial mesh of the CFD domain.

Starting from this grid, an automatic refinement based on the Volume Fraction (VF) of the liquid phase of N_2O was applied, in order to increase the numerical resolution at the liquid–vapour interface and in the vena contracta region. The *adaptive mesh refinement* criterion uses a threshold on the VF gradient to activate local cell subdivision, yielding the refined mesh shown in Figure 3.11.

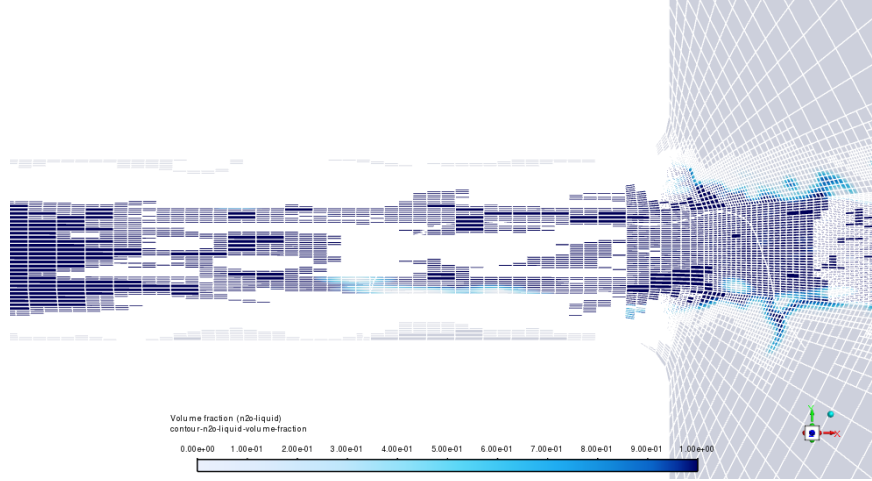


Figure 3.11: Refined mesh with automatic adaptivity.

For the continuous phase, the incompressible VOF model with geometric reconstruction (*Geo-Reconstruct*) of the interface was used. The liquid phase is initialised in the manifold and in the upstream duct, while the gaseous phase fills the downstream chamber. The transition of the atomised jet is modelled through VOF–to–DPM conversion: part of the continuous liquid phase is transferred to a discrete particle phase (unsteady DPM), which evolves in the domain downstream of the orifice.

The boundary conditions follow the logic of the previous single–phase simulations, now including the three–dimensional manifold: at the inlet a **Pressure Inlet** with total pressure equal to the operating line pressure is imposed, while at the outlet a **Pressure Outlet** in the chamber with fixed static pressure is used. The walls of the duct and of the orifice are considered adiabatic, with *no-slip* condition. Turbulence is modelled with the SST $k-\omega$ model, chosen for its robustness in the presence of strong velocity gradients and possible local separations in the vena contracta region. The computation was performed in transient regime until a quasi–steady condition of the mixture flow rate at the outlet was reached. First–order implicit schemes in time and second–order upwind schemes for momentum and turbulence were used, consistently with the choices adopted in the single–phase simulations.

3.2.2 Two–phase CFD results

The resulting flow field structure is summarised in Figure 3.12, which shows the magnitude of the mixture velocity in the orifice region and in the downstream chamber.

The formation of the vena contracta just downstream of the hole is observed, followed by a rapid expansion and a progressive relaxation towards a more diffused two-phase jet.

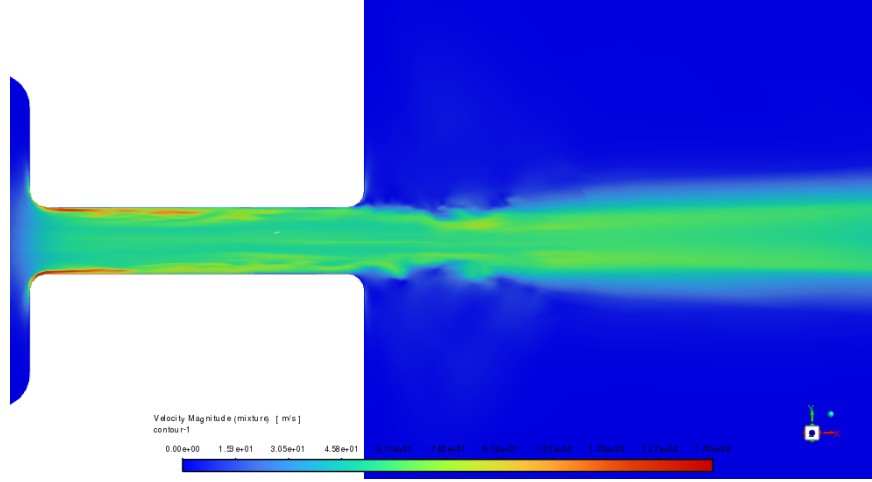


Figure 3.12: Magnitude of the mixture velocity.

The distribution of the Volume Fraction of liquid N_2O along the domain is reported in Figure 3.13. Upstream of the orifice the flow is almost entirely liquid, while downstream the void fraction increases significantly due to the pressure relaxation and vapour formation. The jet shows a central region still liquid, surrounded by two-phase structures and zones with prevailing gas, consistent with a controlled flashing regime. The mass transfer from the continuous VOF phase to the discrete DPM phase generates a cloud of droplets downstream of the orifice, represented in Figures 3.14a–b.

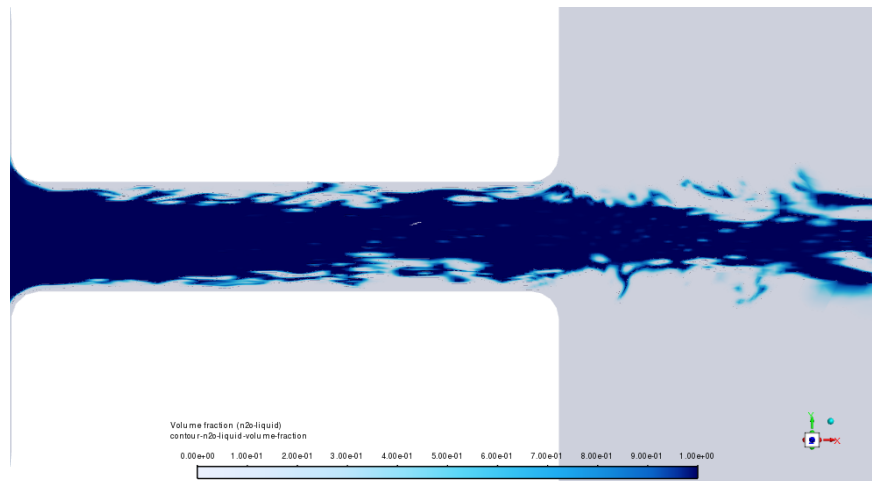
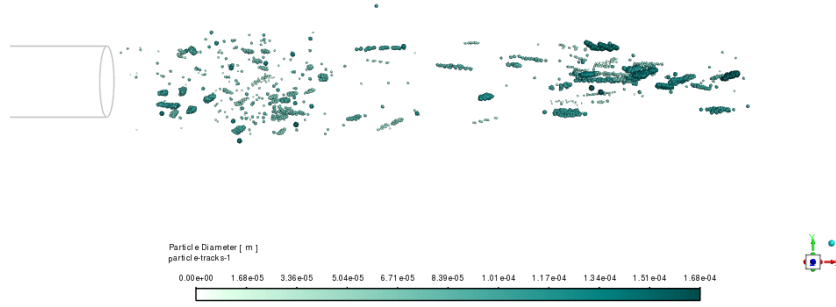
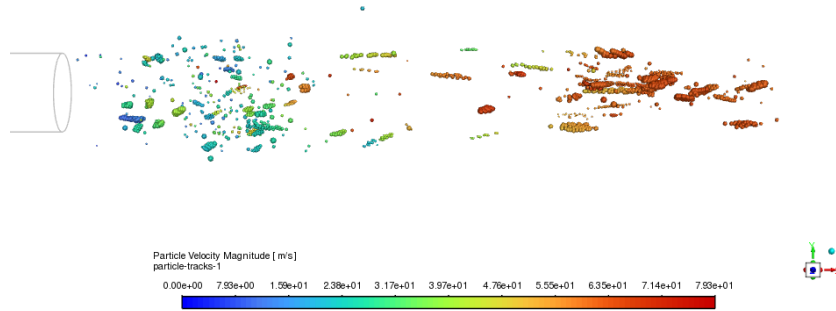


Figure 3.13: Volume Fraction of N_2O (liquid+vapour mixture).



(a) DPM tracks coloured as a function of particle diameter.



(b) DPM tracks coloured as a function of particle velocity.

Figure 3.14: Evolution of the DPM particulate generated by the VOF-to-DPM model.

The total mixture mass flow rate and the average liquid volume fraction were evaluated on a control plane downstream of the orifice (plane p2), for which the following values are obtained:

$$\dot{m}_{\text{CFD}} = 0.1103 \text{ kg/s}, \quad \alpha_{\ell, \text{CFD}} = 0.682.$$

The latter value indicates that, at the outlet section considered for the validation, the flow is clearly two-phase but still dominated by the liquid phase (about 68% in volume), thus configuring an intermediate regime between the nearly single-phase cases previously analysed and a fully vaporised jet.

Figure 3.15 shows the Fluent report for the mixture mass flow rate and the liquid-phase volume fraction at the outlet.

mixture	
Mass Flow Rate	[kg/s]
plane-p2	-0.11032653

(a) Fluent report of mixture mass flow rate on the outlet plane p2.

n2o-liquid	
Area-Weighted Average	
plane-p2	0.68237895

(b) Volume Fraction of the pure liquid phase of N₂O.

Figure 3.15: Report and definition at the outlet section.

3.2.3 Determination of the discharge coefficient and comparison with NHNE

For the comparison with the analytical model, a two-step procedure was adopted. In the first step, the `PyInjection_performance` module was run by setting the discharge coefficient $C_d = 1.0$ and using as boundary conditions the pressures measured directly from CFD at the orifice inlet and outlet:

$$P_{1,\text{CFD}} = 53.68 \text{ bar}, \quad P_{2,\text{CFD}} = 42.27 \text{ bar}, \quad T_{\text{line}} = 293 \text{ K},$$

with orifice diameter $D = 2 \text{ mm}$ and a single hole ($N_h = 1$).

Under these conditions, the output table of the `performance` module, shown in Figure 3.16, provides the following total mass flow rates:

$$\dot{m}_{\text{SPI}}^{(C_d=1)} = 0.134 \text{ kg/s}, \quad \dot{m}_{\text{HEM}}^{(C_d=1)} = 0.100 \text{ kg/s}, \quad \dot{m}_{\text{NHNE}}^{(C_d=1)} = 0.118 \text{ kg/s},$$

with automatic selection of the NHNE model by the *phase-aware* selector.

INPUT PARAMETERS	
Parameter	Value
Fluid	NitrousOxide
T_line [K]	293.000
P1 [bar]	53.680
P2 [bar]	42.269
Orifice D_hole [m]	2.000e-03
D_ref [m] (equiv.)	2.000e-03
Orifice L [m]	-
L/D [-]	-
Edge radius r [m]	-
r/D [-]	-
Nh [-]	1
A_h [m^2]	3.142e-06
A_tot [m^2]	3.142e-06
Cd [-]	1.000
Cd mode	user
Cd source	user-provided Cd
Keep A_tot const (Nh)	OFF
SPI compressible	OFF
n SPI	1.200

MASS FLOW TABLE - TOTAL mdot (SPI / HEM / NHNE / Phase-aware)	
Model	mdot_total [kg/s]
SPI	0.134
HEM	0.100
NHNE	0.118
Phase-aware	0.118

OUTLET PROPERTIES (per hole)		
Phase	liquid	vapor
T [K]	285.451	285.451
T_sat [K]	285.451	285.451
U [m/s]	41.068	26.319
rho [kg/m3]	838.113	123.323
mu [Pa-s]	3.000e-04	1.850e-05
Re [-]	2.295*10^5	3.509*10^5
alpha [-]	0.609	0.391
Cp [J/kg/K]	2668.342	2278.023
MW [kg/kmol]	44.013	44.013
h [J/kg]	1.942e+05	3.919e+05

MAIN RESULTS - TOTAL mass flow and mixture properties	
Parameter	Value
mdot_total [kg/s]	0.118
Phase	two-phase
T [K]	285.451
T_sat [K]	285.451
U_mix [m/s]	67.387
rho_mix [kg/m3]	558.943
mu_mix [Pa-s]	2.757e-04
Re_mix [-]	2.732*10^5
Mach [-]	0.329
sigma [N/m]	0.003

Figure 3.16: Output of the performance module with $C_d = 1$.

Comparing these values with the CFD mass flow rate $\dot{m}_{\text{CFD}} = 0.1103 \text{ kg/s}$, it is observed that:

- the SPI model overpredicts the mass flow rate by more than 20%, confirming that the single-phase limit is no longer representative in this regime;
- the HEM model, constrained to a fully homogeneous and strongly dissipative two-phase equilibrium, underpredicts the mass flow rate by about 10%;
- the NHNE model, which interpolates between SPI and HEM as a function of the distance from the saturation curve, provides an intermediate prediction with an error of about 7%.

In the second step of the validation procedure, the effective discharge coefficient $C_{d,\text{CFD}}$ was derived by comparing the mass flow rate obtained from the VOF-to-DPM simulation with the mass flow rate predicted by the NHNE model in the limiting case $C_d = 1$. Since in the analytical model the NHNE mass flow rate depends linearly on the discharge coefficient, one can write:

$$C_{d,\text{CFD}} = \frac{\dot{m}_{\text{CFD}}}{\dot{m}_{\text{NHNE}}^{(C_d=1)}}.$$

From the CFD simulation, the following value was obtained:

$$\dot{m}_{\text{CFD}} = 0.11032653 \text{ kg/s},$$

while the NHNE model with $C_d = 1$ provides:

$$\dot{m}_{\text{NHNE}}^{(C_d=1)} = 0.118 \text{ kg/s}.$$

The effective discharge coefficient therefore results:

$$C_{d,\text{CFD}} = \frac{0.11032653}{0.118} \simeq 0.935.$$

This value represents the global discharge coefficient of the orifice in the two-phase regime obtained from the VOF-to-DPM simulation.

By inserting the estimated value $C_{d,\text{CFD}} = 0.935$ into the `PyInjection_performance` module, the NHNE model returns a mass flow rate

$$\dot{m}_{\text{NHNE}}^{(C_d=C_{d,\text{CFD}})} \simeq 0.111 \text{ kg/s},$$

consistent with the CFD mass flow rate within numerical errors. Figure 3.17 reports the inputs used and the complete model results, including the mixture state parameters and the mass flow rates predicted by the four analytical models.

INPUT PARAMETERS		OUTLET PROPERTIES (per hole)		
Parameter	Value	Phase	liquid	vapor
Fluid	NitrousOxide	T [K]	285.451	285.451
T_line [K]	293.000	T_sat [K]	285.451	285.451
P1 [bar]	53.680	U [m/s]	38.399	24.608
P2 [bar]	42.269	rho [kg/m3]	838.113	123.323
Orifice D_hole [m]	2.000e-03	mu [Pa.s]	3.000e-04	1.850e-05
D_ref [m] (equiv.)	2.000e-03	Re [-]	2.145*10^5	3.281*10^5
Orifice L [m]	-	alpha [-]	0.609	0.391
L/D [-]	-	Cp [J/kg/K]	2668.342	2278.023
Edge radius r [m]	-	MW [kg/kmol]	44.013	44.013
r/D [-]	-	h [J/kg]	1.942e+05	3.919e+05
Nh [-]	1			
A_h [m^2]	3.142e-06			
A_tot [m^2]	3.142e-06			
Cd [-]	0.935			
Cd mode	user			
Cd source	user-provided Cd			
Keep A_tot const (Nh)	OFF			
SPI compressible	OFF			
n SPI	1.200			

MASS FLOW TABLE - TOTAL mdot (SPI / HEM / NHNE / Phase-aware)	
Model	mdot_total [kg/s]
SPI	0.125
HEM	0.094
NHNE	0.111
Phase-aware	0.111

MAIN RESULTS - TOTAL mass flow and mixture properties	
Parameter	Value
mdot_total [kg/s]	0.111
Phase	two-phase
T [K]	285.451
T_sat [K]	285.451
U_mix [m/s]	63.007
rho_mix [kg/m3]	558.943
mu_mix [Pa.s]	2.757e-04
Re_mix [-]	2.554*10^5
Mach [-]	0.308
sigma [N/m]	0.003

Figure 3.17: Output of the `performance` module with $C_d = 0.935$.

The NHNE model also provides the complete reconstruction of the outlet state, including the properties of the liquid and vapour phases and the mixed quantities. In particular, for $C_d = 0.935$ the following are obtained:

$$T = 285.45 \text{ K}, \quad U_{\text{mix}} = 63.007 \text{ m/s}, \quad \rho_{\text{mix}} = 558.94 \text{ kg/m}^3, \\ \mu_{\text{mix}} = 2.75 \times 10^{-4} \text{ Pa}\cdot\text{s}, \quad \text{Re}_{\text{mix}} = 2.554 \times 10^5, \quad \sigma = 0.003.$$

These values are consistent with the CFD solution, which shows a mixture with high liquid density, a moderate vapour content and a velocity comparable to that predicted by the model.

The internal table of the `performance` module also confirms the relative behaviour of the models:

Table 3.5: Total mass flow rate predicted by each model and by CFD.

Model	$\dot{m}_{\text{total}}$ [kg/s]
SPI	0.125
HEM	0.094
NHNE	0.111
Phase-aware	0.111
CFD	0.1103

The two-phase simulation therefore makes it possible to correctly identify the operating regime as "two-phase", and the *phase-aware* selector automatically activates

the NHNE model, which represents a realistic compromise between the single-phase SPI limit and the strongly dissipative two-phase HEM limit.

3.3 Validation of the semi-empirical estimate of C_d

To complete the analysis, the validation of the discharge coefficient predicted by the semi-empirical model implemented in `PyInjection` is considered. The correlation used combines three contributions: the ISO 5167 (Reader-Harris/Gallagher) formulation developed for sharp-edged orifices, a corrective gain due to the inlet radius r/D , and an additional dissipative term based on the Darcy-Weisbach coefficient. Although physically grounded, this structure has been calibrated on a wide experimental dataset referring to water-air and not to N_2O , and it deliberately incorporates a conservative character to account for geometric uncertainties and the unavoidable viscous losses in short orifices.

In the case analysed here, the discharge coefficient estimated by the geometric model is significantly lower than the value obtained from the single-phase CFD simulation. This difference is expected and can be explained by various factors: the different reference fluid of the original dataset, the lack of a specific calibration for N_2O in cryogenic conditions, and above all the loss models deliberately introduced to obtain a conservative estimate, suitable for the preliminary design phase.

It should also be emphasised that the CFD simulation does not include physical effects that are fundamental for a more rigorous reproduction of the real injector behaviour: gas compressibility, a real-gas model for N_2O , evaporation of the liquid and of the discrete DPM phase. A fully thermo-fluid-dynamic simulation including such effects would be necessary for an accurate validation of the real discharge coefficient, and a dedicated experimental campaign (*cold-flow* or static tests) would likewise be advisable to definitively define the reference value of C_d .

Despite these limitations, the obtained semi-empirical estimate represents a conservative bound that is useful in the preliminary sizing phase, since it tends to underpredict the mass flow rate with respect to the actual value. This behaviour is advantageous during the design stage, reducing the risk of overpredicting the mass flow rate under real operating conditions.

Figure 3.18 shows the output of the `PyInjection_performance` module in the considered case, with the geometrically estimated C_d and the mass flow rates predicted by the analytical models.

INPUT PARAMETERS		OUTLET PROPERTIES (per hole)		
Parameter	Value	Phase	liquid	vapor
Fluid	NitrousOxide			
T_line [K]	293.000	T [K]	285.451	285.451
P1 [bar]	53.680	T_sat [K]	285.451	285.451
P2 [bar]	42.269	U [m/s]	30.302	19.419
Orifice D_hole [m]	2.000e-03	rho [kg/m3]	838.113	123.323
D_ref [m] (equiv.)	2.000e-03	mu [Pa.s]	3.000e-04	1.850e-05
Orifice L [m]	1.000e-02	Re [-]	1.693*10 ⁵	2.589*10 ⁵
L/D [-]	5.000	alpha [-]	0.609	0.391
Edge radius r [m]	5.000e-04	Cp [J/kg/K]	2668.342	2278.023
r/D [-]	0.250	MW [kg/kmol]	44.013	44.013
Nh [-]	1	h [J/kg]	1.942e+05	3.919e+05
A_h [m ²]	3.142e-06			
A_tot [m ²]	3.142e-06			
Cd [-]	0.738			
Cd mode	geom			
Cd source	0.738			
Keep A_tot const (Nh)	OFF			
SPI compressible	OFF			
n SPI	1.200			

MASS FLOW TABLE - TOTAL mdot (SPI / HEM / NHNE / Phase-aware)	
Model	mdot_total [kg/s]
SPI	0.099
HEM	0.074
NHNE	0.087
Phase-aware	0.087

MAIN RESULTS - TOTAL mass flow and mixture properties	
Parameter	Value
mdot_total [kg/s]	0.087
Phase	two-phase
T [K]	285.451
T_sat [K]	285.451
U_mix [m/s]	49.722
rho_mix [kg/m3]	558.943
mu_mix [Pa.s]	2.757e-04
Re_mix [-]	2.016*10 ⁵
Mach [-]	0.243
sigma [N/m]	0.003

Figure 3.18: Results of the PyInjection_performance model with discharge coefficient estimated through the semi-empirical model.

4. Future developments and recommendations

The simulations carried out in this work have made it possible to validate the analytical model both in the liquid single-phase limit and in incompressible two-phase conditions through a VOF-to-DPM procedure. This approach has allowed the behaviour predicted by the NHNE model to be compared with a numerical description consistent with phase separation and vapour evolution in the orifice, representing a first step towards a more complete CFD modelling of the phenomenon. Further developments are nevertheless required to describe more faithfully the rapid expansion of nitrous oxide and the thermodynamic processes that govern its behaviour in flashing conditions. A natural extension would consist in integrating the effects of gas-phase compressibility by activating the energy equation and using a real-gas model for N_2O , so as to correctly represent the enthalpy drop and the thermal effects associated with instantaneous vaporisation. In parallel, enabling evaporation both of the continuous liquid phase (through models compatible with VOF and EWF) and of the discrete phase in the DPM, with the addition of a breakup model such as KHRT, would allow the fragmentation of the film and the formation of droplets to be described more realistically. Such a simulation would provide an even more robust comparison for the validation of the NHNE model and would make it possible to assess more accurately the phase diagnosis implemented in `PyInjection`. A further experimental verification through static tests and *cold-flow* campaigns also remains strongly recommended, as it is essential to quantitatively consolidate the discharge coefficients and the flow-rate predictions in the various operating regimes. In parallel, another development concerns the extension of the analytical model to swirl injectors. The implementation of a dedicated model for the swirler, based on the same philosophy adopted in this report, would make it possible to describe the contribution of pressure loss, swirl generation and *liquid sheet* formation. This model can be validated through CFD simulations employing VOF-EWF-DPM transition models, so as to capture the evolution of the centrifuged film and the subsequent breakup, replicating the numerical structure used for the plain orifice but adapting it to the characteristic phenomenology of swirl injection systems. In this way, the `PyInjection` framework can be progressively extended towards a unified and consistent description of the main injection systems for N_2O , while maintaining strong consistency between analytical model, CFD simulations and experimental validation.

4. Conclusions

The work presented has enabled the development and validation of a comprehensive analytical model for predicting the mass flow rate through orifices fed with N_2O , integrating single-phase and two-phase descriptions through the SPI, HEM and NHNE models and introducing a *phase-aware* selector capable of automatically identifying the operating regime. The CFD simulations performed, both in fully single-phase conditions and via VOF-to-DPM procedures for the two-phase description, have confirmed the reliability of the model in the liquid limit and provided a first qualitative verification of the behaviour predicted near saturation.

The **PyInjection** platform, composed of the **core**, **performance** and **design** modules, now represents a modular and flexible tool for the preliminary analysis of injectors, capable of integrating geometric, thermodynamic and fluid-dynamic aspects in a single framework. The results obtained constitute a solid basis for the future development of more advanced models including compressibility, real-gas thermodynamics and complete evaporation of N_2O , as well as for the extension towards swirl configurations and experimental validation through static and *cold-flow* tests. Overall, the work demonstrates that a combined analytical-numerical approach can provide an in-depth and predictive understanding of the behaviour of short-orifice injectors, laying the groundwork for future optimisation of the feed system and for direct integration into propulsion design processes.

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